

Notebook-Pipeline User Study

Hello! Thank you for your interest in our research. We study how to automatically extract pipelines from computational notebooks to ease the comprehension and migration of underlying data science process in the notebooks. Before proceeding, please read and agree with the following electronic consent.

* Indicates required question

We plan to include the results of this survey in a scientific publication.

When the research results are published or presented, no personally identifiable information will be shared. Your confidentiality will be kept to the degree permitted by the technology used.

If you have any concerns or questions about your rights as a participant or about the way the study is being conducted, please contact nb2p@proton.me

1. Electronic Consent *

Please select your choice below. Selecting the Yes option indicates that: 1) you have read and understood the above information, ii) you voluntarily agree to participate, and iii) you are at least 18 years old. If you do not wish to participate in the research study, please decline participation by selecting No. Please note that this survey will be best displayed on a laptop or desktop computer. Some features may be less compatible for use on a mobile device.

Mark only one oval.

☐ Yes

☐ No

Section 1: Evaluating Ground Truth Construction

In this section, there are 10 notebooks. You will be presented a notebook one at a time. Please read *both the Markdown text and the code* in it and answer the given questions.

NOTE: You do not have to understand the code in all details. It would be enough if you have a basic understanding what is happening there. For example, you see that some lines are about data downloading, others then train a model, and finally, graphs are built based on some lines of code.

Evaluating Ground Truth Construction: Notebook 1

Now, please open: <A_URL_TO_NOTEBOOK>

2. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

[illegible]

3. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

[illegible]

Evaluating Ground Truth Construction: Notebook 2

Now, please open: <A_URL_TO_NOTEBOOK>

4. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

[illegible]

5. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Evaluating Ground Truth Construction: Notebook 3

Now, please open: <A_URL_TO_NOTEBOOK>

6. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly representative

7. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Evaluating Ground Truth Construction: Notebook 4

Now, please open: <A_URL_TO_NOTEBOOK>

8. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

[illegible]

9. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

[illegible]

Evaluating Ground Truth Construction: Notebook 5

Now, please open: <A_URL_TO_NOTEBOOK>

10. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

[illegible]

11. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Evaluating Ground Truth Construction: Notebook 6

Now, please open: <A_URL_TO_NOTEBOOK>

12. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly representative

13. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Evaluating Ground Truth Construction: Notebook 7

Now, please open: <A_URL_TO_NOTEBOOK>

17. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Evaluating Ground Truth Construction: Notebook 9

Now, please open: <A_URL_TO_NOTEBOOK>

18. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly representative

19. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Evaluating Ground Truth Construction: Notebook 10

Now, please open: <A_URL_TO_NOTEBOOK>

20. Alignment of Ground Truth with Workflow Stages *

On a scale of 1 to 10, how well do the Markdown in the notebook represent distinct stages in a typical data science workflow?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly representative

21. Comprehensiveness of Ground Truth *

On a scale of 1 to 10, how comprehensively do the Markdown cover all key stages in the data science workflow presented in the notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Fully comprehensive

Section 2: Evaluating Output Quality of Extracted Segments

In this section, there are 5 Python scripts and a set of generated pipelines from them. You will be presented one script with 4 corresponding pipelines at a time. Each of the script and pipelines are packed in a single notebook for your convenience.

Please read the code in the first code cell that contains the Python script, check the rendered pipeline graph 1-4, each indicated by the title, and answer the given questions.

NOTE: You do not have to understand the code in all details. It would be enough if you have a basic understanding what is happening there. For example, you see that some lines are about data downloading, others then train a model, and finally, graphs are built based on some lines of code.

Evaluating Output Quality of Extracted Segments: Notebook 1

Now, please open: <A_URL_TO_NOTEBOOK>

Python script 1

Now, look at the code cell titled "Python Script".

22. Ease of Code Understanding *

On a scale of 1 to 10, how long do you need to understand the basic workflow of this script? As an example, suppose "Shortest Time" means 5 seconds, and "Longest Time" means 5 minutes. You may provide the answer based on your own time range.

Mark only one oval.

[illegible]

Pipeline 1

Now, look at the two images titled "Pipeline 1" that represent the corresponding pipeline without and with code.

23. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

[illegible]

24. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

[illegible]

25. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

[illegible]

Pipeline 2

Now, look at the two images titled "Pipeline 2" that represent the corresponding pipeline without and with code.

26. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

[illegible]

27. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

1 2 3 4 5 6 7 8 9 10

No improvement Significant improvement

34. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Evaluating Output Quality of Extracted Segments: Notebook 2

Now, please open: <A_URL_TO_NOTEBOOK>

Python script 1

Now, look at the code cell titled "Python Script".

35. Ease of Code Understanding *

On a scale of 1 to 10, how long do you need to understand the basic workflow of this script? As an example, suppose "Shortest Time" means 5 seconds, and "Longest Time" means 5 minutes. You may provide the answer based on your own time range.

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Shor	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Longest Time

Pipeline 1

Now, look at the two images titled "Pipeline 1" that represent the corresponding pipeline without and with code.

36. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

37. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

38. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Pipeline 2

Now, look at the two images titled "Pipeline 2" that represent the corresponding pipeline without and with code.

39. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

40. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

41. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Pipeline 3

Now, look at the two images titled "Pipeline 3" that represent the corresponding pipeline without and with code.

42. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

43. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

44. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Pipeline 4

Now, look at the two images titled "Pipeline 4" that represent the corresponding pipeline without and with code.

45. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

46. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

47. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Evaluating Output Quality of Extracted Segments: Notebook 3

Now, please open: <A_URL_TO_NOTEBOOK>

Python script 1

Now, look at the code cell titled "Python Script".

48. Ease of Code Understanding *

On a scale of 1 to 10, how long do you need to understand the basic workflow of this script? As an example, suppose "Shortest Time" means 5 seconds, and "Longest Time" means 5 minutes. You may provide the answer based on your own time range.

Mark only one oval.

[illegible]

Pipeline 1

Now, look at the two images titled "Pipeline 1" that represent the corresponding pipeline without and with code.

49. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

[illegible]

50. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

[illegible]

57. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

[illegible]

Pipeline 4

Now, look at the two images titled "Pipeline 4" that represent the corresponding pipeline without and with code.

58. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

[illegible]

59. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

[illegible]

60. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Evaluating Output Quality of Extracted Segments: Notebook 4

Now, please open: <A_URL_TO_NOTEBOOK>

Python script 1

Now, look at the code cell titled "Python Script".

61. Ease of Code Understanding *

On a scale of 1 to 10, how long do you need to understand the basic workflow of this script? As an example, suppose "Shortest Time" means 5 seconds, and "Longest Time" means 5 minutes. You may provide the answer based on your own time range.

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Shor	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Longest Time

Pipeline 1

Now, look at the two images titled "Pipeline 1" that represent the corresponding pipeline without and with code.

62. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

63. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

64. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Pipeline 2

Now, look at the two images titled "Pipeline 2" that represent the corresponding pipeline without and with code.

65. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

66. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

67. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Pipeline 3

Now, look at the two images titled "Pipeline 3" that represent the corresponding pipeline without and with code.

68. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

69. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

70. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Pipeline 4

Now, look at the two images titled "Pipeline 4" that represent the corresponding pipeline without and with code.

71. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Perfectly accurate

72. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
No i	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Significant improvement

73. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Evaluating Output Quality of Extracted Segments: Notebook 5

Now, please open: <A_URL_TO_NOTEBOOK>

Python script 1

Now, look at the code cell titled "Python Script".

74. Ease of Code Understanding *

On a scale of 1 to 10, how long do you need to understand the basic workflow of this script? As an example, suppose "Shortest Time" means 5 seconds, and "Longest Time" means 5 minutes. You may provide the answer based on your own time range.

Mark only one oval.

[illegible]

Pipeline 1

Now, look at the two images titled "Pipeline 1" that represent the corresponding pipeline without and with code.

75. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

[illegible]

76. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

[illegible]

80. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

[illegible]

Pipeline 3

Now, look at the two images titled "Pipeline 3" that represent the corresponding pipeline without and with code.

81. Accuracy of Extracted Segments *

On a scale of 1 to 10, how accurately do the extracted pipeline segments match the stages you would expect in a data science workflow for this notebook?

Mark only one oval.

[illegible]

82. Usefulness for Understanding *

On a scale of 1 to 10, how much do the extracted pipeline segments improve your understanding of the notebook's code?

Mark only one oval.

[illegible]

86. Practical Utility *

On a scale of 1 to 10, how useful would you find the extracted pipeline segments in your daily work in data science?

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Extremely useful

Thank you!

You may submit your answers now.

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